

DELHI POLICE

STANDING ORDER NO. 433/2014

Subject: **BUSINESS AREA WATCH SCHEME : DELHI POLICE**

Spread over an area of 1483 sq. kms. with a population of over 1.8 crores, Delhi is amongst the most populous cities in the world with a density of approximately 6000 persons per sq. km. Its growth since independence has outpaced planned development and a major part of city's population lives in slums and areas lacking civic amenities. Consequently, urban anonymity coupled with economic inequalities are among the major crimogenic factors.

Policing in Delhi is an onerous and ever expanding responsibility. However, increase in police strength have not kept pace with this mega city's growth. Consequentially, the policemen cannot always be present at all places. This, however, can be compensated through enhancing community policing and by appropriate usage of technology. Delhi Police has taken a number of community policing initiatives like the Neighbourhood Watch Scheme, Eyes and Ears Scheme, Parivartan, Pehchan, Yuva, Senior Citizens Scheme, tenant/servant verification etc. In all these schemes, the design and format has been provided by the Police, in which the community members participate in policing with Delhi Police acting as facilitator. In the field of technology, Delhi Police has inter-alia been using CCTV coverage of important market places and border checkpoints for prevention and detection of crimes. So far CCTV system has been funded by the government and is being implemented and maintained by ECIL/Delhi Police.

Since requirement of CCTV cameras in the city is very large and police budgetary provisions may not enable coverage of all vulnerable places in one go, alternate financing models may become necessary. In this regard, installation of CCTV network/s with public assistance is relevant and can be suitably implemented to meet Delhi's expanding needs.

In order to further augment the CCTV coverage in the city, it has been decided to initiate a **Business Area Watch Scheme** as community initiative with Delhi Police acting as the facilitator of the scheme and public spirited members of the community funding the installation, maintenance and operation of CCTV converge in an area, particularly markets and other commercial/institutional locations.

This Business Area Watch Scheme (BAWS) of Delhi Police will be as follows:-

1. It will be implemented in all the districts in a phased manner.
2. All District Addl. CsP/DCsP will immediately identify vulnerable areas from crime or terror point of view, where the market or business community is willing to participate by way of installation of CCTV cameras and its maintenance under police guidance.
3. The Local Monitoring Centre of these cameras will be set up at the respective BAWS areas and, if required, the cameras will be connected to Cyber Highway by linking them with concerned Police Station. The locations chosen should generally be such that linking upto the Police Station is not cost prohibitive. It may be noted that the Cyber Highway connectivity is already provided to all the Police Stations in Delhi. In addition, a wide area network setup by ECIL exists at market places where the government funded CCTV system is already installed.
4. The desired minimum configuration of the CCTV Cameras, linking network, storage, monitoring etc. will be as formulated by Delhi Police. The same is given at **Annexure**. The Market/Business Community will be required to adhere to these standards. The concerned community members can decide the number of cameras, area to be covered etc. based on cost factor.
5. Once such an area is identified/shortlisted, the Addl.CP/DCP concerned will hold a meeting with the concerned market community members to explain the concept, cost/effort involved and benefits of the scheme. Addl.CsP/DCsP District shall suitably motivate the community members to willingly participate in the scheme after understanding its advantages

and costs. The BAWS will be managed by a five to ten member committee comprising of the persons owning/running establishments in the area. The President of the market/area association or a person selected by the association shall be the Chief Coordinator. He will be assisted by other public persons from the area who will be designated as Co-ordinators. One of them shall be nominated as treasurer. The concerned Jt.CP Range/Addl.CP/DCP will act as Mentors of the scheme and will closely monitor the implementation of the scheme. The concerned SHO and the Division Officer will also be the ex-officio Co-ordinators of the BAWS. In order to ensure smooth functioning of the BAWS, a monthly meeting of Coordinators will be held at a date and time fixed by the Chief Coordinator in consultation with the SHO and other Coordinators. A minute book will be maintained for proper recording of the decisions taken during monthly meetings.

6. The number of cameras and coverage area will be decided by the community members in consultation with Delhi Police to effectively meet the security needs of the area.
7. Connectivity will be done preferably through optical fibre.
8. CCTV cameras may be a mix of fixed cameras, mega-pixel fixed cameras and PTZ cameras as per requirement. Option for advanced facilities like Licence Plate Readers (LPR) and Video Analytics may also be incorporated, where necessary.
9. The BAWS shall hire staff for round-the-clock monitoring of CCTV display at the Local Monitoring Centre and for safe-keeping of equipments. Storage will also be done at this centre. The beat-staff and senior officials of the Police Station shall regularly visit the area of the Business Area Watch Scheme and oversee the functioning of the monitoring centre. Where space for the monitoring centre is an issue, the police may help in providing requisite accommodation at police booth/kiosk etc.
10. The networking/connectivity from individual cameras upto the Local Monitoring Centre will be extended upto the Police Stations, if required. A monitoring station should be made at the Police Station also, if not already existing. In case the ECIL CCTV monitoring station exists in the Police Station, the feed of

the BAWS cameras should come upto this station and should be integrated with the ECIL system. From the Police Station, the feed of the BAWS cameras will be able to ride the cyber-highway of Delhi Police so that the same can be monitored centrally in C4i or any other alternate control facility.

11. Since the cost of the system is to be borne by the community, there should be no undue interference by police with regard to the mechanism adopted for procurement of hardware, software and maintenance personnel by the BAWS. Endeavour should be to only ensure that minimum technical standards are met and the system is properly maintained. The BAWS could, however, be advised to raise funds by voluntary contributions and devise a transparent system of approvals for expenditure and maintenance of accounts.

Bhim Sain Bassi

12/8/2014

(BHIM SAIN BASSI)
COMMISSIONER OF POLICE;
DELHI

Order Book No. 15 /HAR/PHQ 12/8/2014

No. 3287-3536/HAR/PHQ dated Delhi, the 12-8-2014

Copy forwarded to the:-

1. All Special Commissioners of Police, Delhi.
2. Managing Director, Delhi Police Housing Corporation.
3. All Joint Commissioner of Police and Additional Commissioner of Police, Delhi/New Delhi.
4. Principal, P.T.C., Jharoda Kalan, Delhi.
5. S.O. TO C.P., Delhi.
6. All Distts./Units DCsP including FRRO, New Delhi.
7. All ACsP/PHQ.
8. PA to C.P., Delhi.
9. All ACsP Sub Division (through concerned Addl. CsP/DCsP).
10. All SHOs (through concerned Addl. CsP/DCsP).
11. All Inspectors/PHQ.
12. Library/PHQ.
13. HAR/PHQ with 10 spare copies.

ANNEXURE

CCTV Fixed Camera Technical Specifications

S/no	Specification	Description
1	Camera Type	Day/Night.
2	Image Sensor	1/3" interline progressive CCD
3	Active Pixels	752X582
4	Horizontal Resolution	540 TVL
5	SENSITIVITY	Colour 0.2 lux at 30 IRE, B/W 0.05lux at 30 IRE
6	SNR	>= 50 DB
7	AGC	ON/Off Selectable
8	Synchronisation	Internal, Line Lock
9	Electronic Shutter	1/50s - 1/100000s or better
10	Back Light Compensation	ON/Off Selectable
11	Video Output	1V(P-P),75 Ohms
12	Video Compression	MPEG4/H.264
13	Video Resolution	4 CIF at 25 fps
14	Video Streams	Dual Streaming/Ip Configurable
15	Supported Protocols	TCP/IP,UDP/IP,HTTP,RTP(All selectable)
16	Web Browser	Enabled
17	Lens Mount	C/CS
18	Lens Focal Length	To be decided
19	Operating Voltage	24VAC or 12 VDC
20	Power Consumption	<5Watts
21	OPERATING TEMPERATURE	(0)° C TO 50° C
22	Storage Temperature	(0)° C TO 60° C
23	Network Interface	10/100 Base -T, POE
24	Environmental protection	IP 66 and Vandal proof.

CCTV PTZ Camera Technical Specifications

Sno	Specification	Description
1	Camera Type	Day/Night. HR dome camera
2	Image Sensor	1/4" progressive CCD
3	Active Pixels	752X582
4	Horizontal Resolution	540 TVL
5	OPTICAL ZOOM	36X
6	DIGITAL ZOOM	12X
6	SENSITIVITY	Colour 0.05 lux at 30 IRE, B/W 0.005lux at 30 IRE
7	LENS	3.4 - 122.4 @ F1.6
8	SNR	>= 50 DB
8	AGC	ON/Off Selectable
9	Synchronisation	Internal, Line Lock
10	Electronic Shutter	1/50s - 1/10000s or better
10	Back Light Compensation	ON/Off Selectable
11	Electronic Image Stabilization	Supported
12	PAN Speed	1 to 100 degree
12	Panning Range	360 degree endless
13	Tilt Range	90 degree
14	Tilt Angle	1-120 degree/s
14	Presets	99 presets or better
15	Video Output	1V(P-P),75 Ohms
16	Video Compression	MPEG4/H.264
16	Video Resolution	4 CIF at 25 fps
17	Video Streams	Dual Streaming
18	Supported Protocols	TCP/IP,UDP/IP,HTTP,RTP,DNS,NTP(All selectable)
18	Web Browser	Enabled
19	Controls	On Screen Camera status
20	Camera Title	16 characters user definable or above
20	Alarm IN/OUT	Minimum 2 each
21	OPERATING TEMPERATURE	(0)° C TO 50° C
22	Storage Temperature	(0)° C TO 60° C
22	Network Interface	10/100 Base -T, POE +
23	Environmental protection	IP 66 and Vandal proof.

CCTV Mega Pixel Camera Technical Specifications

S/no	Specification	Description
1	Camera Type	Day/Night.
2	Image Sensor	1/2.7" progressive CMOS
3	Active Pixels	2MP or 3MP or 5MP
4	SENSITIVITY	Colour 0.2 lux at 30 IRE, B/W 0.05lux at 30 IRE
5	SNR	>=50 DB
6	Electronic Shutter	1 - 1/10000s or better
7	Back Light Compensation	ON/Off Selectable
8	Video Output	1V(P-P),75 Ohms
9	Video Compression	MPEG4/H.264
10	Frame Rate	9FPS for 5MP / 15FPS for 3MP / 20FPS for 2MP
11	Video Streams	H.264, MPEG4, dual streaming
12	Supported Protocols	TCP/IP,UDP/IP,HTTP,RTP(All selectable)
13	Web Browser	Enabled
14	Lens Mount	C/CS
15	Operating Voltage	24VAC or 12 VDC
16	OPERATING TEMPERATURE	(0)° C TO 50° C
17	Storage Temperature	(0)° C TO 60° C
18	Network Interface	10/100 Base -T, POE
19	Environmental protection	IP 66 and Vandal proof.

LPR CAMERA TECHNICAL SPECIFICATION

Sl No	Specification	Description
1	Image sensor	Progressive scan CMOS
2	Lens	Varifocal 10-40 mm CS-mount and DC-iris
3	Day and Night	Automatically removable infrared-cut filter
4	Minimum illumination	1 lux (Color), B/W: 0.3 lux or better, F1.2
5	Shutter Time	1/24500 s to 1/2 s
6	Resolution	1280x800 , and other resolutions below and above
7	Frame Rate	30 fps in all resolutions
8	Video Compression	H.264, MPEG-4, Motion JPEG
9	Video Streaming	Multiple, individually configurable streams in H.264 and Motion JPEG Controllable frame rate and bandwidth VBR/CBR H.264
10	Image Settings	Compression, color, brightness, sharpness, contrast, white balance, exposure control, backlight compensation, fine tuning of behavior at low light, rotation Text and image overlay, privacy mask
11	Field of view for Licence Plate recognition	up to 3.5 meters
12	Supported Network Protocols	IPv4/v6, HTTP, HTTPS, QoS Layer 3, SNMPv1/v2c/v3(MIB-II), NTP, RTSP, TCP, UDP, IGMP, ICMP, DHCP, ARP.
13	Maximum Installation height supports	Upto 6.5 Meters
14	IR Power On	Sensor Auto Control, Adjustable Control Point
15	IR Distance	10 to 20 meters
16	IP Rating	IP 66

Video Management Software Specifications

The Video Management software should perform administration and configuration database of entire CCTV Surveillance system.

The software performs the following applications

- Creates Database for all cameras
- Add or Remove the cameras
- Creates the live viewing profile and recording profile
- Allows to create rules for recording
- Creating users and user groups and permissions
- Patterns for PTZ Cameras
- Diagnostics for the system and cameras

The Video Management System consists of the following major software components,

- Recording and Management Server
- Client Viewer: For viewing real-time video data or recorded video data Matrix Monitor
- Event Management: For supporting graphical alarm management.

VMS Client Specifications:

The client should performs the following applications simultaneously without interfering with any of the Archive Server operations (Recording, Alarms, etc.):

- Live display of cameras
- Live display of camera sequences
- Control of PTZ cameras
- Playback of archived video
- Retrieval of archived video
- Instant Replay of live video
- Configuration of system settings
- Execution of system macros
- View camera in a video tile 2 X 2 , 4 X 4 etc.
- View Map or procedure in a video tile
- Starting/stopping PTZ pattern

Technical Specifications for VMS Server

Sino	Specification	Description
1	CPU	Single-INTEL Quad core,5540 sreies 2.5 GHz or above ,8mb internal cache per socket 2 Processor capable
2	Memory	16Gb,DDR-III,800/1066MHz ECC memory, Upgradable to 64 GB
3	HDD Controller	On board support for raid 0,1 and 5 on sas controller.
4	HDD	2*300GB SAS H/S HDDs. Support at last 14 HDDs
5	Ethernet ports	2* Dual port GIGA bit NICcards with autosensing .shall have ISCSI
6	PCI slots	provision for 2PCI express PCIx4 or PCIx8
7	Graphics adopter	Standard Video adopter with 8 MB /16MB.
8	Optical drive	DVD R/W. internal drive
9	form factor	rack model with rail kit
10	I/O ports	2* USB Front, 2* usb rear 1* serial, 1VGA port.
11	Reduntant power supply	Reduntant power supply .1+1.
12	Mangement	Remote mangement facility IPML-version 2 compliant
13	OS	certified for windows server 2003 /2008 and linux
14	Bench mark	Specin2006 :above 25: specfp_rate2006:Above 160. Server should have published specint and specfp results

Technical Specifications of External Storage

Sno	Specifications	Description
1	Host Interface	Storage system with iSCSI on 1 Gbit / sec. on Copper.
2	Protocol support	Support for Block and File serving over Ethernet networks.(CISF & NFS Block level)
3	Memory	Cache memory 8GB.
4	Usable Storage capacity	1 TB HD with SATA -II h / s, HDDs arranged in DAE units, Provision for expansion. Maximum capacity can be achieved upto 42TB
5	RAID level Support	0, 1, 5, 6& 10 with redundant Dual RAID Controllers with Dual ported.
6	Logical Units (LUNS)	Realized with 1 TB capacity SATA-II h / s HDDs with global hot spare.
7	Ethernet connectivity	2 x Dual Gigabit Ethernet connectivity for no single point failover.
8	Form factor	Rack Model
9	Power supply & Fans	Redundant power supply & Redundant Cooling Fans
10	Host OS support	Certified for Windows server 2008 or Linux or better

Client Workstation Technical Specification

SI No	Specification	Description
1	CPU	INTEL Xeon dual core W 3503 2.4 GHz or above , 4MB L2 cache
2	Memory	2GB DDR3-ECC 1066MHzFSB
3	Graphics adapter	Dual 256 MB PCIE X16 Nvidia nvs 295
4	Ethernet ports	Integrated dual giga bit ethernet ports autosensing
5	Optical drive	16 xDVD R / W drive internal
6	Hdd's	2 x 250 GB SATA -drives
7	HDD Controller	SATA internal support for RAID 1
8	Form factor	Tower model
9	Keyboard	Standard key board same as OEM
10	Mouse	Standard optical scroll mouse same as OEM
11	I/O ports	2* USB, 1* serial.
12	Monitor	17 TFT colour monitor
13	Audio	Speaker and mike set
14	OS	Windows 7 professional or better

42"LED BACK LIGHT MONITOR Technical Specifications

SN O	SPECIFICATIONS	Description
1	RESOLUTION	1920 X 1080p
2	DISPLAY TYPE	FULL HD,EDGE LED BACKLIGHT
3	SCREEN SIZE	42(107cmS)
4	VIEWING ANGLE (DEGREE)	178/178
5	ASPECT RATIO	16:09
6	BRIGHTNESS	440 CD/M ² OR BETTER
7	RESPONSE TIME (GTG)	8ms or BETTER
8	BROADCASTING SYSTEM	PAL/SECAM -BG/ DK, PAL-1,NTSC-M
9	VIDEO INTERFACE	HDMI/DISPLAY PORT
10	AUDIO INPUT (FOR VIDEO)	AV1/2/3:RCA Phono type connectors X1 (1set side,2sets rear)
11	POWER SUPPLY	AC100-240 Vac 50/60Hz
12	STAND BY MODE	1W

24"LED BACK LIGHT MONITOR Technical Specifications

SN O	SPECIFICATIONS	Descriptpion
1	SCREEN SIZE	24"
2	DISPLAY TYPE	FULL HD,EDGE LED BACKLIGHT
3	MAX.RESOLUTION	1920 X 1080P
4	BRIGHTNESS	400cd/m ²
5	ASPECT RATIO	16:09
6	VIEWING ANGLE(H/V)	160°/160°
7	DISPLAY COLOR	16.7 million
8	RESPONSE TIME	< 5ms (typ.)
9	VIDEO INTERFACE	HDMI/DISPLAY PORT
10	CONSUMPTION	<110W
11	ELETRICAL RATINGS	AC100~240V 50/60Hz

5KVA UPS Technical Specifications

Sno	Specification	Description
1	Capacity in VA	5KVA,3500Watts
2	Technology	Microprocess controlled double conversion On line
3	Self test	UPS should be capable of performin self test during power on to ensure reliable operation
4	INPUT	
a)	Voltage range	Wide range from 160v to 280 v on full load, 100v to 280 V on half load
b)	Frequency range	45 to 55 hz
c)	Phase	Single phase / three phase with ground
5	Power factor	Active power factor correction>0.99
6	OUTPUT	
a)	Voltage	230VAC 1Ph / 415VAC 3-PH
b)	Voltage Regulation	±1%
7	Harmonic Distortion(THD)	< 2% (linear load) < 3% (non-linear load)
8	Output wave form	Pure sine wave
9	Crest Factor	> =3:1
10	Efficiency AC-DC	High efficiency> 92%
11	Power factor	0.7 O/P=3500watts
12	Charger	Built-in solid state float/ trickle charger modes with current limiting features
13	Transfer Time	0ms
14	ColdStart	Available
15	User Friendly front panel	UPS should have display of informative LED indicators along with Mimic display of all important parametres
16	Audible alarm	mains failure alarm low battery alarm and overload
17	Protection	Advanced Electronic protection for device safety backed with MCBs/MCCBs /fast acting fuses.high speed pulse blanking, Electronic over voltage/under voltage.
18	Constrution	<=3U / Rack Mountable Version Tower type
19	Standard	RS 232 port for software interface
20	Battery configuration	Min VAH-3456 for 30 mins back up on full load
21	Standards	Mandatory ISO 9001,ISO 14001.Restriction of Hazardous Substances (RoHS),VDE.Preferred GOST,C-tick,En50091-1,EN50091-2,EN55022ClassA,EN60950,EN61000-3-2
22	Warranty	UPS-Three years &Batteries for two years.

10KVA UPS Technical Specifications

Sno	Specification	Description
1	Capacity in VA	10KVA, 8000Watts
2	Technology	Microprocess controlled double conversion On line
3	Self test	UPS should be capable of performin self test during power on to ensure reliable operation
4	INPUT	
a)	Voltage range	3 Ph-wide range from 277V to 284V on fullload,
b)	Frequency range	45 to 55 hz
c)	Phase	Single phase / 3-phase with ground
5	Power factor	Active power factor correction >0.99
6	OUTPUT	
a)	Voltage	230VAC 1Ph / 415VAC 3-PH
b)	Voltage Regulation	±1%
7	Harmonic Distortion(THD)	<2% (linear load)<3% (non-linear load)
8	Output wave form	Pure sine wave
9	Crest Factor	>=3:1
10	Efficiency AC-DC	High efficiency> 92%
11	Power factor	0.8 O/P=8000watts
12	Charger	Built-in solid state float/ trickle charger modes with current limiting features
13	Transfer Time	0 ms
14	ColdStart	Available
15	User Friendly front panel	With Mimic display of all important parametres
16	Audible alarm	Mains failure alarm low battery alarm and overload
17	Protection	Advanced Electronic protection for device safety backed with MCBs/MCCBs /fast acting fuses.high speed pulse blanking, Electronic over voltage/under voltage
18	Constrution	<=6U / Rack Mountable Version Tower type
19	Standard	SNMP with HTTP,HTTPS,SMTP,SNMP,SSL,TCP/IP,Telnet,WAP.RS 232 port for software interface
20	Battery configuration	Min VAH -6912 for 30 mins back up on full load
21	Standards	Mandatory ISO 9001,ISO 14001.Restriction of Hazardous Substances (RoHS),VDE.Preferred GOST,C-tick,En50091-1,EN50091-2,EN55022ClassA,EN60950,EN61000-3-2
22	Warranty	UPS-THREE years & Batteries for TWO years.

15KVA UPS Technical Specifications

Sno	Specification	Description
1	Capacity in VA	15KVA, 12000Watts
2	Technology	Microprocess controlled double conversion On line
3	Self test	UPS should be capable of performin self test during power on to ensure reliable operation.
4	INPUT	
a)	Voltage range	3 Ph-wide range from 277V to 284V on fullload,
b)	Frequency range	45 to 55 hz
c)	Phase	Single phase/3-phase with ground
5	Power factor	Active power factor correction >0.99
6	OUTPUT	
a)	Voltage	230VAC 1Ph / 415VAC 3-PH
b)	Voltage Regulation	±1%
7	Harmonic Distortion(THD)	<2% (linear load)<3% (non-linear load)
8	Output wave form	Pure sine wave
9	Crest Factor	≥3:1
10	Efficiency AC-DC	High efficiency> 94%
11	Power factor	0.8 O/P=12000watts
12	Charger	Built-in solid state float/ trickle charger modes with current limiting features
13	Transfer Time	0 ms
14	ColdStart	Available
15	User Friendly front panel	UPS should have display of informative LCD indicators along with Mimic display of all important parametres.
16	Audible alarm	Mains failure alarm low battery alarm and overload
17	Protection	Advanced Electronic protection for device safety backed with MCBs/MCCBs /fast acting fuses.high speed pulse blanking, Electronic over voltage/under voltage.
18	Constrution	6U / Rack Mountable Version Tower type
19	Standard	SNMP with HTTP,HTTPS,SMTP,SNMP,SSL,TCP/IP,Telnet,WAP.RS 232 port for software interface
20	Battery configuration	Min vah -16128 for 30 mins back up on full load
21	Standards	EN /EC 6240,EN/EC 62040-1-1
22	Warranty	UPS-THREE years & Batteries for TWO years.

Industrial Grade Fieldswitch Specifications

Sno	Specification	Description
Hardware & Performance Requirements		
1	Architecture	Fixed Configuration Layer 2 Din-rail mountable Switch.
2	Switching Capacity	3.2Gbps of Switching Capacity.
3	Switch Forwarding Rates (Layer 2) & buffer size	Switching (Layer 2) forwarding rate of 3Mpps and Packet Buffer Size of 1 Mb
4	No. of Ports	6/8 nos. 10/100 Base TX ports and 2-Nos of 100Base FX SC ports.
5	Noise Control	broadcast/multicast/uni-cast noise control to prevent degradation of switch performance from faulty end stations
Layer 2 Functionality		
6	Type of VLANs	Port based VLANs
7	No. of VLAN	4 active VLANs.
8	IGMP	IGMPv2 / IGMPv3 snooping
Management		
9	Terminal	Telnet,console and web Connectivity
10	Network monitoring/management	SNMP v1, v2 and v3, ,SSHv2, telnet and web management
11	Port Mirroring	For monitoring network traffic of a particular port/VLAN/group of ports/entire switch.
Environmental Specifications		
12	Operating Temperature:	0 to 75 degree c
13	Humidity	5 to 95% (non-condensing)
14	Housing	Metal, IP30 protection

Fieldswitch Specifications

Sno	Specification	Description
Hardware & Performance Requirements		
1	Architecture	Fixed Configuration Layer 2 Rack mountable Switch with support for redundant power supply.
2	Switching Capacity	8.8Gbps of Switching Capacity.
3	Switch Forwarding Rates (Layer 2) & buffer size	Switching (Layer 2) forwarding rate of 6Mpps, Packet Buffer Size of 1 MB
4	No. of Ports	4 nos. 10/100 Base TX ports and 2 or 4 Nos of SFP / SFP combo ports.
5	Noise Control	Support for broadcast/multicast/uni-cast noise control to prevent degradation of switch performance from faulty end stations
Layer 2 Functionality		
6	Type of VLANs	port based VLANs
7	VLAN Trunking Protocol or Equivalent	VTP or equivalent protocol to reduce administrative burden of configuring VLANs on multiple switches
8	No. of VLAN	8 active VLANs.
9	IGMP	IGMPv2 / IGMPv3 snooping
Management		
10	Terminal	Telnet, web and console Connectivity
11	Network monitoring/management	Syslog, SNMP v1, v2 and v3, ,SSHv2, telnet and web management
12	Port Mirroring	For monitoring network traffic of a particular port/VLAN/group of ports/entire switch.
Software		
13	Software Version	Version of software for supplied switch would be latest release to support all required features

Core Switch Specifications

Sno	Specification	Description
Hardware & Performance Requirements		
1	Architecture	Fixed Configuration Multilayer and Rackmountable Switch with Rackmount kit and additional supports for redundant power supply
2	Switching Fabric Capacity	96 Gbps of Switching Fabric Capacity.
3	Routing Capability (Layer 3)	RTP, OSPF and dynamic routing protocols
4	Port Specifications	20 * SFP ports + 4 Nos 10/100/1000 SFP Combo ports, The SFP Ports support for 1000Lx and 100Fx Transceivers
Connectivity & Filtering		
5	802.3ad	Industry Standard Port/Link Aggregation for All Ports
6	Noise Control	broadcast/multicast/unicast noise control to prevent degradation of switch performance from faulty end stations
Layer 2 Functionality		
7	Type of VLANs	Port and subnet based VLANs
8	VLAN Trunking Protocol or Equivalent	VTP or equivalent protocol to reduce administrative burden of configuring VLANs on multiple switches
9	No. of VLAN	1000 active VLANs
10	MAC Addresses	12000 MAC addresses or more
Layer 3 Functionality		
11	L3 Routing Protocol	RIP/RIP2 natively and OSPF, BGP and Multicast routing
12	IPv6	IPv6 routing protocols like RIPng and OSPFv3
Security Features		
13	Access Control Lists	Standard and Extended ACLs (various type of ACLs like port based/time based)
14	Integrated Security	Integrated security features like DHCP snooping, Dynamic Arp Inspection, IP Source guard and uRPF (unicast Reverse path forwarding)
15	MAC Address Filtering	Should Support MAC Address Filtering based on source and destination address
16	802.1X	802.1x Network Security and RADIUS Authentication
Spanning tree Protocols		
17	STP	Spanning Tree Protocol, Rapid Spanning Tree Protocol, Multiple Spanning Tree Protocol
Multicasting		
18	Multicast	H/W or S/W based IPv4 Multicasting
19	IGMP	IGMP v2 and v3
Network Management		
20	Network monitoring/management	The network manageable through NMS on per port/switch basis with common interface for all manageable devices on the network, and supports syslog, SNMP, RMON, SSH, telnet, web management
21	Port Mirroring	Port mirroring feature for monitoring network traffic of a particular port/VLAN/group of ports/entire switch.
IEEE Standards Compliance		
22	IEEE Standards	Ethernet: IEEE 802.3z, IEEE 802.3ab, 100Base-T, 1000BASE-T, 1000BASE-X (mini-GBIC/SFP), 1000BASE-SX, 1000BASE-LX/LH, 1000Base-ZX and CWDM based SFP support
		IEEE 802.1D 1S & 1W for Spanning-Tree Protocol
		IEEE 802.1p CoS Prioritization
		IEEE 802.3x Flow Control
		IEEE 802.3ad Link Aggregation

Servers Switch Specifications
(Access switch with 1G port for Servers in Control Room)

Sno	Specification	Description
Hardware & Performance Requirements		
1	Architecture	Fixed Configuration Layer 2 Rack mountable Switch, with additional support for redundant power supply
2	Switching Capacity	96Gbps of Switching Capacity.
3	Switch Forwarding Rates (Layer 2)	Switching (Layer 2) forwarding rate of 84Mpps.
4	No. of Ports	20 nos. 10/100/1000 Base TX ports and 4 Nos of 1000SFP / SFP combo ports.
Connectivity & Filtering		
5	802.3ad	Industry Standard Port/Link Aggregation for All Ports
6	Noise Control	broadcast/multicast/uni-cast noise control to prevent degradation of switch performance from faulty end stations
Layer 2 Functionality		
7	Type of VLANs	port based VLANs
8	VLAN Trunking Protocol or Equivalent	VTP or equivalent protocol to reduce administrative burden of configuring VLANs on multiple switches
9	No. of VLAN	8 active VLANs.
Security Features		
10	MAC Address Filtering	MAC Address Filtering based on source and destination address
11	802.1X	802.1x Network Security and Authentication
12	ACL	ACL to filter traffic based on different parameters like TCP ports and on the basis of Time
Multicasting		
13	IGMP	IGMPv2 / IGMPv3 snooping
Management		
14	Terminal	Telnet, console and web Connectivity
15	Network monitoring/management	syslog, SNMP v1, v2 and v3, SSHv2, telnet and web management
16	Port Mirroring	port mirroring feature for monitoring network traffic of a particular port/VLAN/group of ports/entire switch.
Software		
17	Software Version	Version of software for supplied switch would be latest release to support all required features

15U Indoor Rack Technical Specifications

S.No.	Feature/Item	Description
1	Dimensions	Width: 600 mm x Height:15U x Depth: 600 mm
2	Enclosure Construction	Robust steel sheet construction made from 1.5 mm steel sheet, folded and welded from a single piece, with protection channel around the door opening.
3	Doors	2mm sheet steel, with hinges fitted on the left or right for single-door, enclosures, 130° opening angle; with IP 55 protection through IP55 gasketing.
4	Lock System	Suitable locking arrangement with unique key to avoid key duplicity
5	Surface Finish	Nano-ceramic/pure polyester powder coated with RAL 7035 after proper pre-treatment (preferably EC dip coat), with textured Pure Polyester (PP) thickness of 80 to 120 Microns.
6	Mounting	C plinth of 3 mm thick of 100 mm height for easy grouting on to the floor.
7	Cable Entry	PG 29 cable glands the bottom for cable entry: 4 no.'s , Protection category: IP 55 or better
8	19" Mounting	L type 19" angles for mounting the equipments
9	Cable Manager	2 nos. of open horizontal 1U cable manager with 5 nos. of PVC cable loops
10	PDU	2 no. of 19" mountable 1U height power socket strip with 6 nos. of 5A & 15A universal pin sockets with 1.5m mains power cord shall be Provided